MISSION CRITICAL

COSMIC PLAYGROUND

The Final Frontier in Real-Time Physics.
A high-fidelity framework for celestial mechanics, relativistic visualization, and aerospace logistics.

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>> INITIALIZING ENGINE... DONE
>> INTEGRATING RK4... DONE
>> LOADING RELATIVISTIC MESH... DONE
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Cosmic Playground bridges the gap between gaming and academia. Experience a dual-mode 3D simulator where you govern the birth of galaxies or pilot the survival of multi-stage launch vehicles.

01. SPACETIME LAB

The Spacetime Grid

The grid is your primary diagnostic tool. It visualizes the curvature of space using a **Dynamic Vertex Mesh**.

MASS INCREASE
↓
CURVATURE DEEPENS

→

GRID DEFORMATION
↓
GRAVITY WELL

→

ORBITAL TRAP
↓
SATELLITE MOTION

Celestial Inventory & OVS

- **SUN:** G-Class Source
- **BLACK HOLES:** Singularity
- **ASTEROIDS:** Low mass
- **MARS/EARTH:** Habitable
- **NEUTRON STARS:** Dense
- **COMETS:** High eccentricity

Orbital Velocity Scaling (OVS): Precision placement requires vector priming. Click and drag the velocity anchor to project **Future-Path Predictions**. Green lines indicate stable eclipses; Red indicates atmospheric collision.

REALISTIC GRAVITY

Full N-Body. Total Perturbation.

ARCADE GRAVITY

Point-Mass focus. Simplified navigation.

02. THE ROCKET LAB

Aero-Dynamics Chart

Flight success is determined by the interplay of four primary vectors:

THRUST vs. DRAG vs. GRAVITY vs. LIFT

Thermal Load (Re-entry):

Thermal accumulation peaks at 40km altitude. Failure to maintain a consistent **Angle of Attack** will lead to structural ablation.

PRESSURE / TEMP RATIO

AI Launch Coach

Always-on telemetry provides real-time guidance:

- **[0-10KM]**: "Maintain Ascent Profile. Monitor Wind Shear."
- **[MAX-Q]**: "Throttling down to 65% to reduce structural stress."
- **[VACUUM]**: "Stage separation confirmed. Circularizing orbit."

Flight Outcomes

✔ STABLE ORBIT
✔ ESCAPE VELOCITY

✘ THERMAL BURN-UP
✘ PLANETARY IMPACT

03. OPERATIONS

Temporal Time HUD

Time is a scrubbable dimension. The **Forensic Rewind** allows you to backtrack mission failures.

REWIND [-4x] | REAL-TIME [1x] | WARP [64x]

Progression: Mission Logs

Earn **Exploration Score** points to unlock advanced stellar templates.

MASTER

Gravity Slingshot: 500pts

EXPERT

Mars Storm Landing: 750pts

Interface Controls

DRAG: Rotate View
SCROLL: Deep Zoom
PINCH: Radial Orbit
TAB: Toggle Modes
SPACE: Stasis Field
R: Full Reset